Palaeomagnetism of the Ragunda intrusion and dolerite dykes, central Sweden

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The large Ragunda intrusion (1320 Ma) and dolerite dykes cutting the intrusion near its western margin have been extensively sampled for palaeomagnetic study. After comprehensive alternating field and thermal demagnetisation, the intrusion yields a mean direction $D = 20°$, $I = 46°$ ($\alpha_{95} = 7°$); both polarities are present. Two stages of dyke intrusion are identified from their magnetisations: a small number of dykes have directions similar to the Ragunda intrusion and the remainder yield a mean $D = 182°$, $I = 24°$ ($\alpha_{95} = 14°$); both polarities are present in each case and neither dyke suite correlates with the magnetisations of post-Jotnian (ca. 1200 Ma) dolerites from elsewhere in the Fennoscandian Shield. Magnetisations reside in different phases of titanomagnetite, pyrrhotite is a locally important remanence carrier, but maghemite, although sometimes present, has not been identified as a significant carrier. The palaeomagnetic poles help to clarify the form of the apparent polar wander path for the Fennoscandian Shield in Late Proterozoic times.

□ *Palaeomagnetism, thermomagnetism, polar wandering, pole positions, intrusions, dykes, dolerite, gabbro, anorthosite, titanomagnetite, Proterozoic, Fennoscandian Shield, Ragunda, Jämtland, Västernorrland, central Sweden, N6302 N6322 E1650 E1600.*

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The Ragunda intrusion is one of the largest anorogenic Proterozoic intrusions in the Swedish part of the Fennoscandian Shield. Welin & Lundqvist (1975) quoted an age of 1350 Ma and Kornfält (1976) recorded an Rb–Sr age of 1320 ± 30 Ma. Ancillary evidence that it predates the interval of Jotnian sedimentation (Sederholm 1927) is provided by the occurrence of sandstone with Jotnian affinities in fissures in the granite (Kornfält 1969). Together with the Nordingrå intrusion and the Jotnian sandstones and lavas of western Sweden, this body provides one of the major opportunities, within the Swedish part of the Shield, for defining the apparent polar wander (a.p.w.) record between the time of cooling and uplift of the Svecofennian rocks (~1600 Ma) and the intrusion of the post-Jotnian dolerites (~1250 Ma).

Geological setting

The Ragunda intrusion comprises anorthosite and gabbro intruded by a range of granites and syenites and emplaced in Svecofennian metasediments and granites (Högbom 1899; Kornfält 1976). It has structural and petrological affinities with other large Proterozoic bodies intruded into cratonic regions which had already stabilised. These intrusions appear to be practically unique to that part of Middle-Late Proterozoic times between approximately 2000 and 1000 Ma. (Bridgwater & Windley 1973) and relate the range of physical conditions pertaining to this interval (Bridgwater et al. 1973). Högbom (1909) found the dolerite facies confined to the deeper erosion levels and postulated that the intrusion has a layered form 300–400 m in thickness with a core of quartz dolerite/gabbro surrounded by granites and syenites. More recently, Kornfält (1969, 1976) has reported detailed geological and petrological studies. He shows that the laccolithic hypothesis must be rejected because the gabbro facies also occur near the margins of the body and contacts with the country rocks are near-vertical; an origin by magmatic stoping is thus most plausible (Kornfält 1976).

Some parts of the intrusion, notably around the mountain Stadsberget, are extensively cut by dolerite dykes with variable trends between E–W and NW–SE. They are typically 1–2 metres in thickness.

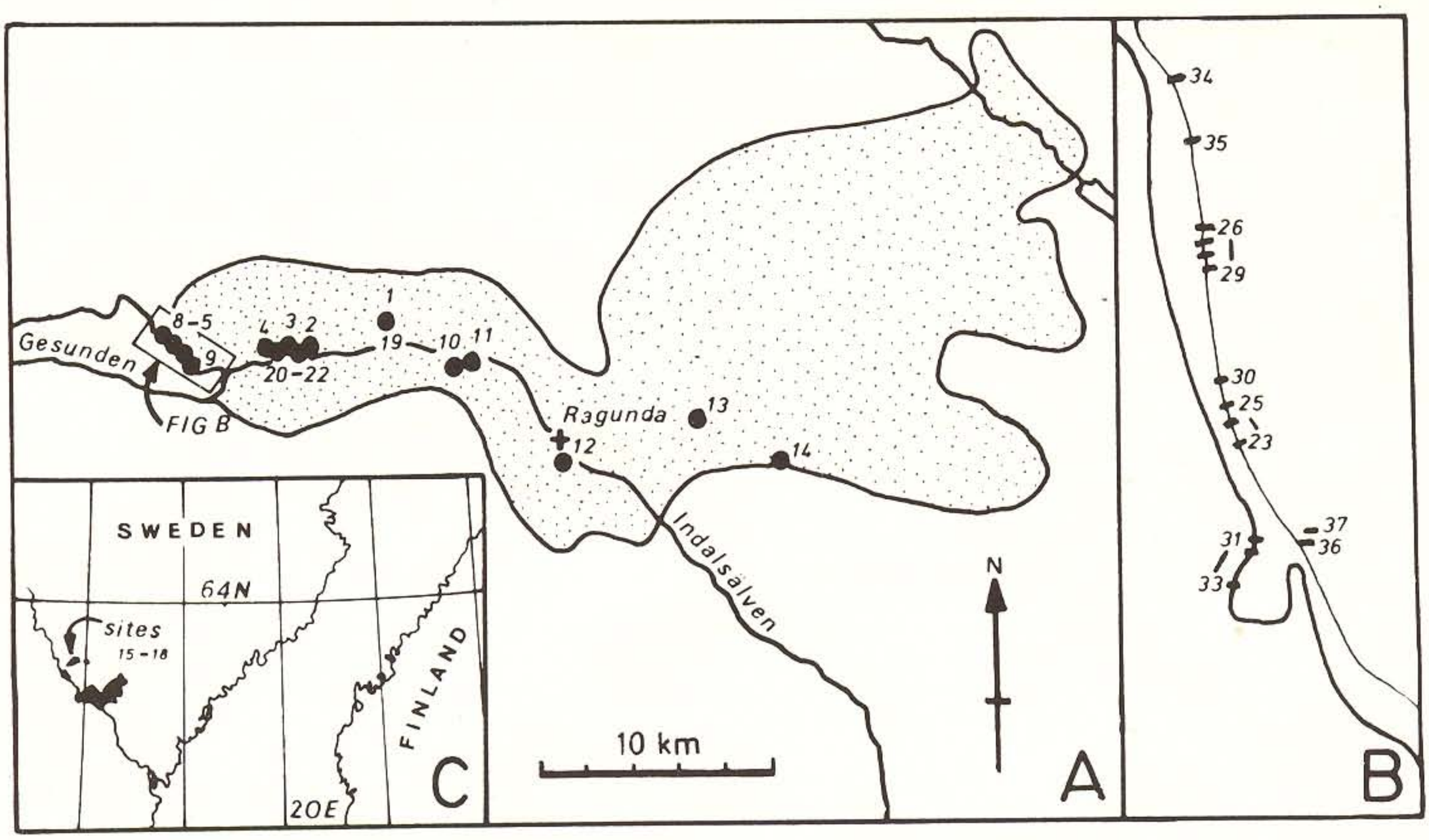


Fig. 1. □ A. Location of palaeomagnetic sampling sites within the outcrop of the Ragunda Rapakivi-type intrusion, central Sweden. □ B. Sampling sites in the dolerite dyke swarm cutting the western part of the intrusion close to the shore of Lake Gesunden. □ C. The regional location of the outcrop of the Ragunda intrusion.

The palaeomagnetic samples

The samples for palaeomagnetic study were drilled in the field using a portable two-stroke motor, and oriented by sights on the sun while still attached to the outcrop. A single site typically comprises seven samples distributed over several metres of outcrop. Much of the area has a heavy drift cover and sampling was concentrated on the well exposed area including road cuts near Lake Gesunden and at the western part of the outcrop of the intrusion (Fig. 1). The sites 1–9 are in intermediate or basic facies of the intrusion and sites 19–37 are in dolerite dykes; sites elsewhere (10–14) have mainly sampled the granitic facies of the intrusion and sites 15–18 are from granite forming the subsidiary intrusion at Borgvattnet.

Palaeomagnetic results

The palaeomagnetic cores were measured in the laboratory using an automated parastatic magnetometer system, and were progressively demagnetised in 500 Hz alternating fields (a.f.) in steps of 5–30 mT (milliTesla) and then in steps of 10 mT to fields of up to 150 mT; the demagnetising procedure was terminated when no further significant or systematic changes in the direction of the natural remanent magnetisation (NRM) were observed. To determine whether or not the directions isolated by the a.f. treatment reside in single, high-blocking temperature, components in titanomagnetite, and relate to the time of initial cooling, one sample from each site was further progressively demagnetised by thermal treatment at temperature intervals up to the Curie point; the incremental steps of 100°C were reduced to 25 or 30° near the Curie point. In addition to resolve the nature of the magnetic remanence carriers, small (ca. 1 mm^3) samples were continuously heated in a saturating magnetic field. The resulting thermomagnetic determinations (saturation magnetisation [J_S] *versus* temperature) allow individual remanence carriers to be recognised from their Curie points.

As anticipated, intensities of magnetisation are closely controlled by chemical composition (Table 1), and are highest in the intermediate and basic facies of the intrusion. The granitic facies have intensities one or two orders of magnitude less, and some material from these sites had to be rejected because it was too weak to measure. The majority of sites exhibit simple and systematic behaviour to progressive a.f. and thermal treatment. Sites 1 and 7 are highly stable and

Table 1. Palaeomagnetic results for sampling sites in the Ragunda Rapakivi-type intrusion (15.9–16.2°E, 63.1–63.5°N).

D and *I* are the mean declination and inclination of the magnetisation calculated from *N* samples with precision parameter *k* (= (*N*–1)/(*N*–*R*)) and circle radius α_{95} degrees about the mean direction at the 95% confidence level. *Dp* and *dm* are the radii of the oval of 95% confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively. Thermomagnetic curves are described as A, heating and cooling curves similar, B, heating curve above cooling curve, C, heating curve below cooling curve, D, large increases in saturation magnetisation on cooling, (x) kink in thermomagnetic curve at 150–200°C. Heating and cooling Curie points are both given where they are different.

Sample locality	*N*	*D*	*I*	*R*	*k*	α_{95}	Virtual geomagnetic pole °N	°E	Range of remanent intensities (M_0) $\times 10^{-6}$ A·m²/kg	Curie point(s) ±15°C	Thermomagnetic curve type
1	7	7.0	33.4	6.96	143	5.1	44.9	187.1	1.48– 5.90	535	A
2	6	13.6	35.2	5.31	7	26.6	45.3	178.1	1.18– 4.67	535	A
3	6	196.8	–55.6	5.99	811	2.4	–61.1	–12.5	0.22– 1.10	535	A
4	6	193.7	–56.1	5.98	274	4.1	–62.3	–7.5	0.90– 4.28	530	A
5[a]	6	112.0	64.2	5.62	13	19.2	31.5	65.7	0.75–10.59	535/520	A
6[a]	5	333.0	75.3	4.99	299	4.4	77.7	–65.1	0.47– 2.29	530–515	A
7	6	26.4	58.7	5.98	262	4.2	61.6	150.3	0.10– 0.30	530	A
8	4	42.0	51.2	3.96	73	10.8	49.2	136.2	0.05– 0.90	310, 500	C
9	7	213.9	–53.2	6.99	439	2.9	–53.9	–35.4	0.31– 1.41	515	A
10	4	28.7	41.6	3.92	39	14.9	46.4	157.0	0.50–17.64	500/485	A
11	6	30.3	59.3	5.92	65	8.4	60.9	144.1	3.81–10.58	535/525	A
12[a]	7	305.9	68.8	5.98	6	27.2	60.1	–69.3	0.53– 1.10	550	A
13	7	201.4	–26.7	6.97	186	4.4	–38.8	–10.5	1.39– 2.46	570	A
14	7	8.3	41.4	6.59	15	16.2	50.3	184.6	0.01– 0.12	570	A
15	8	27.1	38.6	7.86	50	7.9	44.8	159.9	0.01– 0.05	–	–
16	4	27.6	23.0	3.95	63	11.7	35.3	162.8	0.01– 0.05	–	–
17	6	349.9	52.9	5.80	25	13.6	59.7	213.4	0.01– 0.04	540	A
18	3	17.3	48.9	2.77	9	44.5	54.9	169.8	0.00– 0.01	–	–
Mean (15 sites)							Palaeomagnetic pole				
	20.1		45.7	14.5	31	6.9	166.6°E, 51.6°N ($dp = 5.7°$, $dm = 8.9°$)				
Mean (19 sites)[b]											
	20.5		48.7	18.3	27	6.6	164.9°E, 54.0°N ($dp = 5.7°$, $dm = 8.7°$)				

[a] Excluded from overall mean calculation.
[b] Including four dolerite dykes (Table 2).

following initial fall in moment with slight change in direction during the first 5–15 mT treatment, show no further movement, and the stable end points also show no further change with thermal treatment to 530°C (Fig. 2). Sites 2, 5, 8, 10, 11 and 12 yield similar directions but have a lower coercivity NRM's and are less stable to progressive treatment. At sites 5 and 10, stable end points were not achieved in all samples by a.f. demagnetisation and could only be completely recovered by thermal treatment. Notable amongst this group are some samples from site 2 which appear to show a complete double reversal (Fig. 2) during demagnetisation with associated decrease and increase in NRM intensity. It is possible that these samples cooled sufficiently slowly through their critical range of blocking temperatures to record a reversal of the magnetic field.

Sites with magnetisations approximately opposite to this main group are numbers 3, 4, 9 and 13 which include the most highly stable samples of the collection. Over the first few steps of treatment, intensities of magnetisation increase (Fig. 2) as a component in the present Earth's magnetic field is subtracted from the NRM. This is accompanied by a movement away from the present field direction to a southerly negative magnetisation. NRM's isolated from the samples by progressive a.f. treatment are highly stable to subsequent thermal treatment and appear to reside in a single component with a high blocking temperature range (ca. 500–600°C). Site 6 has a stable NRM near to the present Earth's field direction but is highly stable to both a.f. and thermal treatment. A similar mean direction at site 12 listed in Table 1 does not survive thermal treatment and may be a

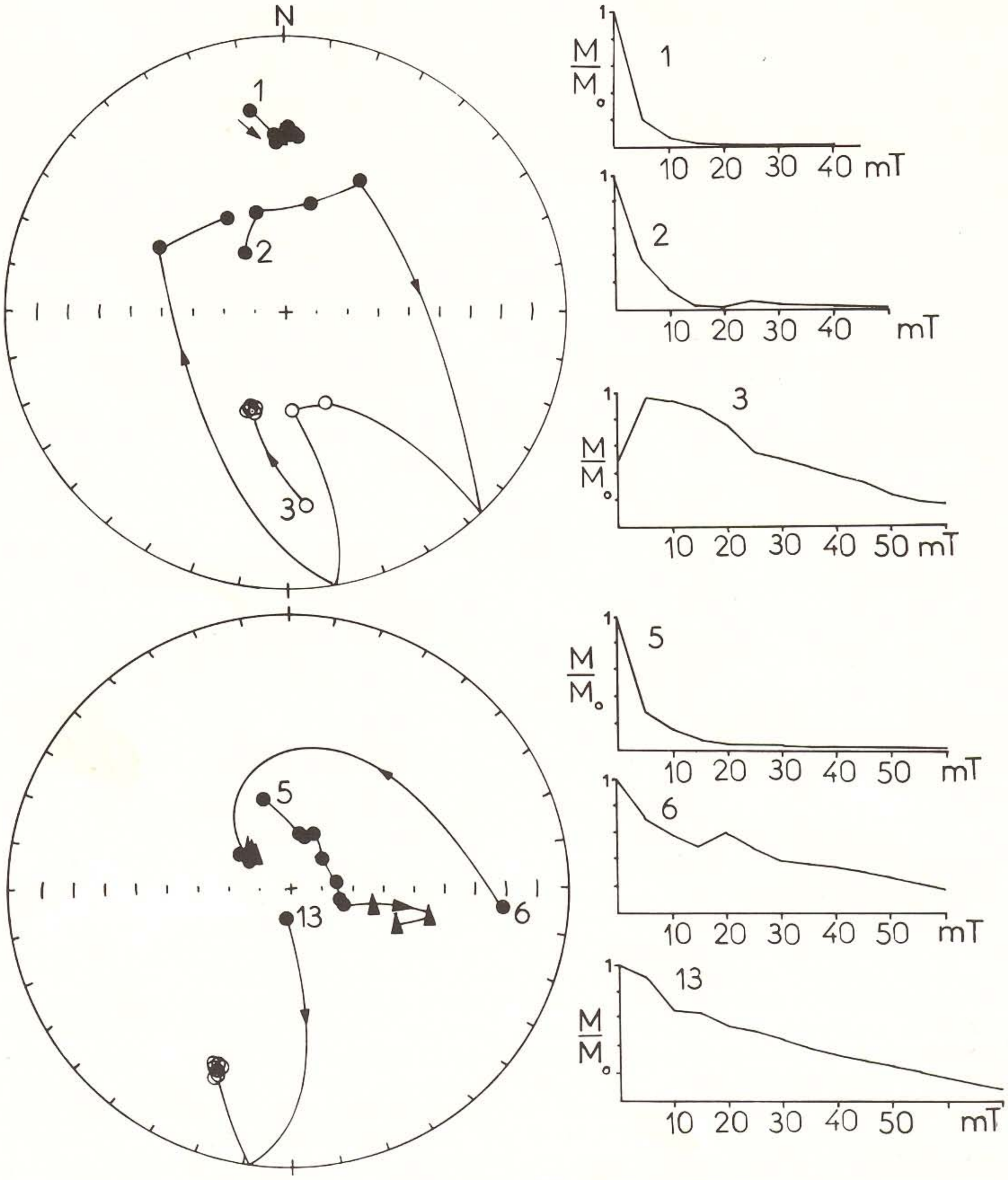


Fig. 2. Behaviour of directions of magnetisation with progressive cleaning treatment for some samples from sites in the Ragunda intrusion. The directions of magnetisation with a.f. treatment are plotted as closed circles (lower hemisphere plots) or open circles (upper hemisphere plots) and with thermal treatment are plotted as triangles (same convention). The projections are stereographic and the decay of magnetic moment (M) with progressive treatment is expressed as a fraction of the initial NRM (M_0).

present field component. Site 5 yields an intermediate direction listed after thermal treatment; this does not correlate with either the northerly positive or southerly negative directions shown by the remaining sites. It seems probable that sites 5 and 6 record a transitional field direction because their location lies between sites 3, 4 and 9 (southerly negative NRM's) and sites 7 and 8 (northerly positive NRM's) of opposite polarity. The weakly-

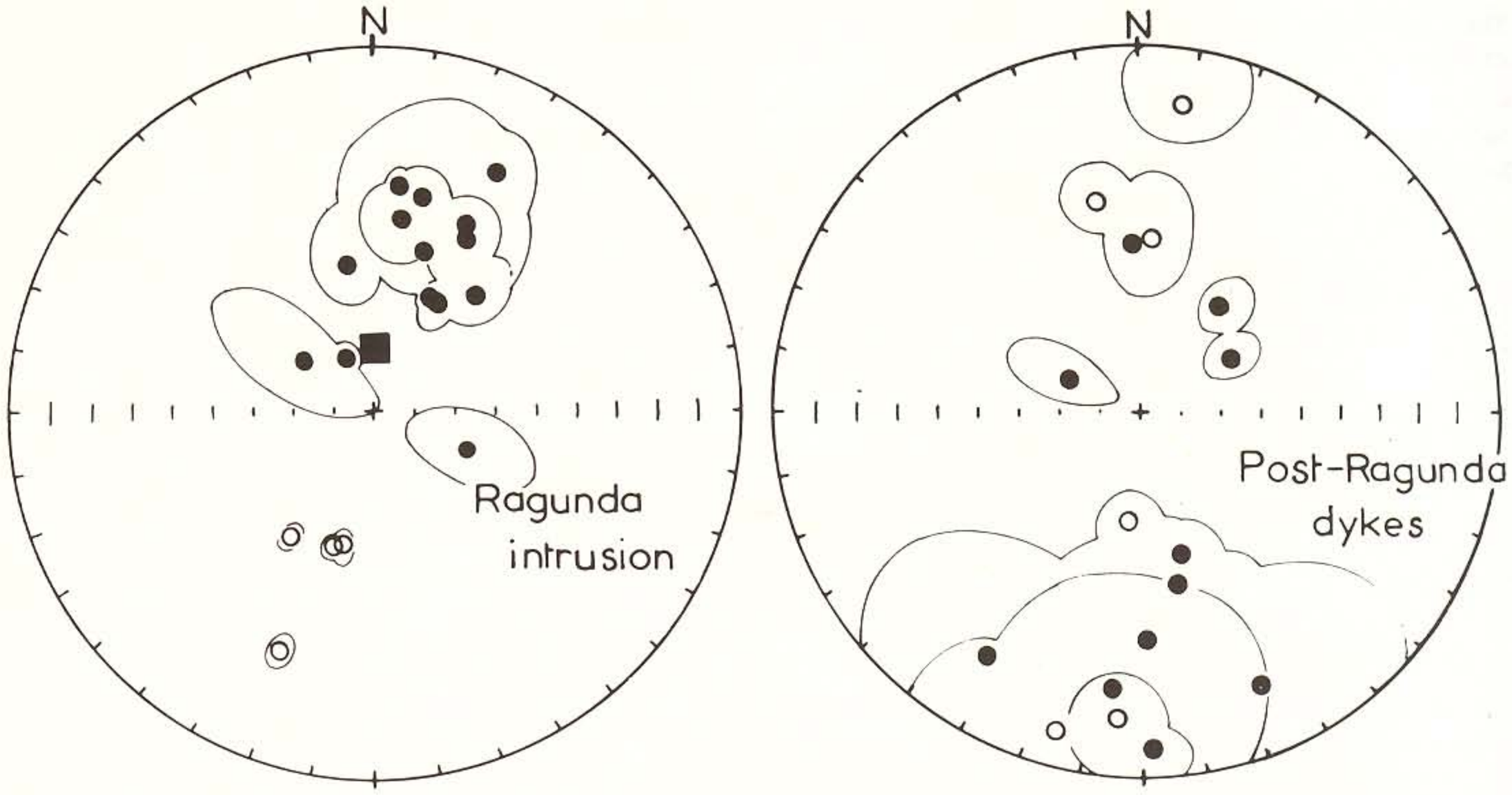


Fig. 3. Site mean directions of magnetisation with 95% confidence circles from the Ragunda intrusion and dykes cutting the intrusion. The square indicates the direction of the present mean dipole field in this area.

magnetised sites 15 to 18 in granites from the north-westerly outcrop yield no meaningful results with a.f. cleaning but give systematic directions of magnetisation with thermal treatment above 500°C. The directions (Table 1) are comparable with the (northerly positive NRM) sites in the main outcrop at Ragunda, supporting the apparent genetic link between the intrusions.

All site mean directions of magnetisation are listed in Table 1 together with Fisher's (1953) statistics, and are plotted in Fig. 3. In general the quality of within-site grouping achieved, as expressed by the precision parameter k, is directly related to coercivity of the NRM and is highest at sites 1 and 7 and the sites with southerly negative magnetisation. It is clear from Fig. 3 that a complete reversal of magnetisation has been detected within the intrusion. Omitting the steep inclination sites 5, 6 and 12 and reversing the sites 3, 4, 9 and 13 yields a combined mean direction of $D=20.1°, I=45.7°$ ($\alpha_{95}=6.9°$) and a palaeomagnetic pole at 167°E, 52°N.

The dolerite dykes (sites 19–37) typically have lower coercivity NRM's than the Ragunda intrusion and correspondingly lower stabilities. This is reflected in the generally poorer within-site groupings of directions after a.f. cleaning (Table 2). All sites except 26 and 32 however, yield significant directions of magnetisation after this treatment. Thermal treatment was only partially successful with this collection and a range of temperatures promoted changes in magnetic susceptibility and increases in magnetic intensity accompanying erratic direction changes. Only the a.f. cleaned directions can accordingly be used, but the thermomagnetic spectra (next section and Table 2) suggest that they are likely to be valid because the remanence resides in magnetite or components with lower blocking temperature ranges.

The dyke mean directions of magnetisation are not simple to interpret because both positive and negative inclinations with northerly and southerly declinations are present (Fig. 3). However, we may note that sites 31, 33, 34 and site 37 (the last one reversed with respect to 31, 33 and 34) are similar to the normal and reversely magnetised directions from the Ragunda intrusion, and all of them have high within-site precisions more typical of the intrusion. These sites are also characterised by A and B type thermomagnetic spectra (next section) which are also typical of samples from the intrusion. The mean direction of magnetisation of these four sites is $D=23°$, $I=60°$ ($\alpha_{95}=19°$) which is not significantly different from that of the main Ragunda intrusion. The remaining sites (with the exception of No. 23) then fall into two groups with northerly and southerly magnetisations (Fig. 3) quite different from the Ragunda body. Sites 25, 28 and 30 are approximately antiparallel to the remainder, and the combined group mean of the 12 sites is $D=182°$, $I=24°$, ($\alpha_{95}=14°$).

This interpretation of the palaeomagnetic data

Table 2. Palaeomagnetic statistics and magnetic properties, post-Ragunda Rapakivi intrusion dykes. The meaning of the symbols is given in Table 1.

Sample locality	*N*	*D*	*I*	*R*	*k*	Range of remanent intensities (M_0) $\times 10^{-6}$ A·m²/kg	Curie point(s) ± 15°c	Thermo-magnetic curve type
19	6	185.1	–14.4	5.88	43	0.14– 4.25	525/480	A
20	5	156.3	16.8	4.58	9	0.01– 0.03	–	D
21	4	178.3	6.1	3.97	87	0.01– 0.03	415	D
22	6	168.0	47.2	5.50	10	1.18– 7.40	525/450	D (x)
23[a]	6	296.4	70.5	5.82	27	3.52–18.29	550/520	A
24	7	164.4	54.3	6.86	44	0.30– 1.94	–	–
25	5	4.5	–46.6	4.90	42	2.61– 1.78	525/515	B
26[a]	3	–	–	–	–	0.00– 0.01	550/500	D
27	6	186.7	21.8	5.05	5	0.00– 0.01	550	D
28	6	8.3	–13.0	5.83	30	0.01– 0.52	535/525	D
29	5	195.2	–7.6	4.21	5	0.01– 0.20	–/535	D
30	6	349.2	–36.0	5.88	42	1.73– 2.90	550	B
31[b]	6	58.3	63.5	5.95	96	1.32– 1.89	550	B (x)
32[a]	6	–	–	–	–	0.58– 2.79	550	A
33[b]	6	37.5	56.1	5.88	42	0.22– 1.05	575	A (x)
34[b]	6	357.2	48.0	5.96	141	2.59– 4.02	575	A
35	4	179.3	34.3	3.59	7	0.01– 0.04	495	D
36	4	213.4	18.6	3.61	8	0.51– 1.05	480	D
37[b]	6	187.3	–63.9	5.91	54	0.40– 1.72	565/540	B
Group means:						*Palaeomagnetic poles*		
(1)	12	181.7	23.5	10.9	10	(α_{95}= 14.3°) 14.4°E, 14.5°S (dp= 8.1°, dm= 15.3°)		
(2)	4	22.6	60.2	3.9	25	(α_{95}= 18.7°) 25.7°E, 64.2°S (dp= 21.5°, dm= 28.4°)		

[a] Sites excluded from group mean calculations.
[b] Sites included in group mean calculation (2).

from the dyke swarm suggests that at least two phases of dyke intrusion are present; one is closely linked to the intrusion of the Ragunda laccolith, and the other defines a quite different palaeofield axis which is presumably considerably younger in age.

Carriers of magnetisation

Microscopic examination of the opaque mineral phases may suggest genetic links between these phases and the character of the magnetisations determined from the laboratory demagnetisation experiments. However, since resolvable phases are much larger than the single or pseudosingle magnetic domains (<0.03 μm) any conclusions drawn from microscopic study are usually of a rather general nature. The morphological character of the larger magnetite grains (>5 μm) appears to be related in no way to stability and this mineral is present at stable sites both as anhedral, partially resorbed, grains (e.g. sites 3 and 7) and as euhedral and skeletal grains (e.g. sites 1 and 6). Magnetic stability is also unrelated to the development of exsolution lamellae within the magnetite grains, in contrast to the correlation widely observed in Mesozoic and Tertiary basalt lavas and dykes (e.g. Ade-Hall et al. 1968), and the only site showing extensive exsolution is number 10 with a relatively poor within-site precision. Maghemite overgrowths around magnetite grains are common, but again appear to be related in no way to magnetic stability; thus this phenomenon is moderate to well developed at very stable sites 3, 5 and 7 and at less stable site 2. The sulphide content however, shows a qualitative correlation with stability, sulphides being abundant at less stable sites such as 2, 5 and 8 and sparse at very stable sites such as 1 and 3, although even here it is not possible to generalise because sulphides are sparse at the less stable sites 10 and 12. Since only the sulphide pyrrhotite ($Fe_{1-x}S$) is ferrimagnetic, and since this mineral (Curie point approximately 300°C) has not been clearly detected on any of the thermomagnetic curves, it is possible that the apparent link between sulphide content and stability reflects indirectly on the nature of co-existing magnetite and is related to the oxygen fugacities of the magmatic liquid

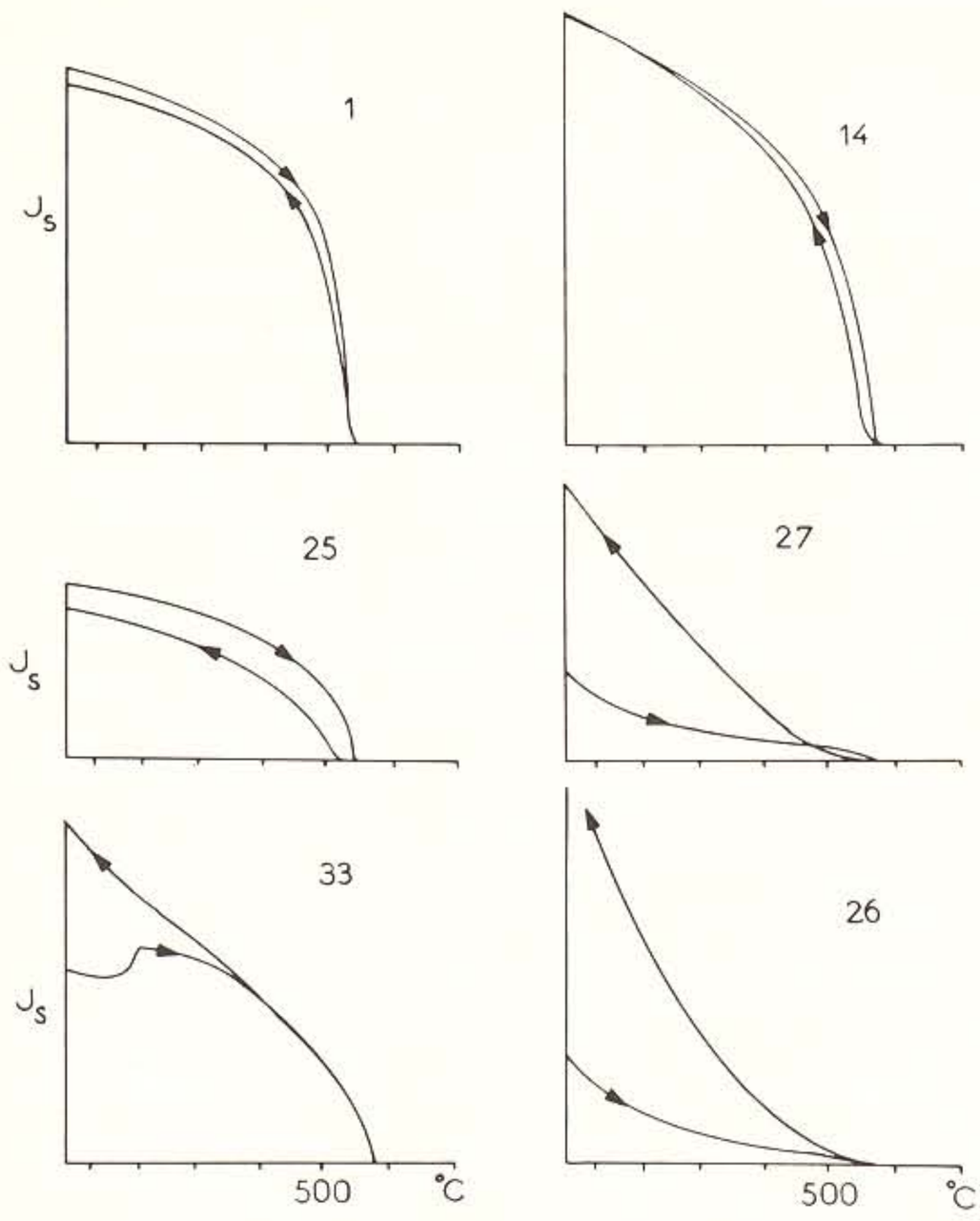


Fig. 4. Thermomagnetic spectra (saturation magnetisation [J_S] *versus* temperature) for the samples of the Ragunda intrusion and dykes cutting the intrusion. The J_S axis is arbitrary and the numbers refer to site numbers.

during initial igneous cooling (see Ade-Hall et al. 1968).

The best clue that can be obtained from microscopic examination about the nature of the stable remanence lies in the development of fine needles of magnetite oriented along cleavage planes in silicates. These are typically up to 1 μm in width and 5 μm in length and go down to the limits of resolution ($\sim$0.5 μm). They are widely distributed at the very stable sites such as 3, 4 and 9, and are either not prominent or not visible at the less stable sites such as 2 and 12. It is difficult to make a quantitative assessment of their importance from microscopic examination alone, but Strangeway et al. (1968) and others have both practically and theoretically demonstrated the importance of such needleshaped grains as stable remanence carriers. Not all the stable remanence can reside in types of grains however, because their properties fail to accommodate stable low coercivity remanence such as that observed at site 1; fine needles of magnetite were not observed at this locality.

The thermomagnetic spectra are very similar for most samples from the intrusion and indicate that titanomagnetite close to pure magnetite in composition is the origin of the single, high-blocking-temperature component (e.g. Fig. 4). The titanomagnetite would appear to have equilibriated during slow igneous cooling because it undergoes little or no oxidation on heating (Type A curves, Table 1) and only four sites (5, 6, 10, 11) show a significant change in Curie point after heating. The exception is site 8 which also yields a low blocking temperature component with a distinct Curie point at 310°C. This component (probably pyrrhotite) undergoes oxidation with increase in J_S on cooling and the curve is irreversible. The maghemite (γFe_2O_3), which is widely distributed as a magnetite overgrowth, does not appear to contribute significantly towards the NRM, because this mineral converts to hematite (αFe_2O_3) on heating, and only at sites 8 and 11 were significant residual moments (about 1% of initial saturation remanence) due to hematite (Curie point 670°C) observed.

No comprehensive microscopic study of the dyke swarm has been attempted, but in general these rocks show quite different thermomagnetic characteristics from the Ragunda intrusion indicating that the remanence carriers have a different nature. Many sites, including those with the poorest within-site groupings (Table 2) show a rapid fall in saturation remanence at low temperatures. Site 20 exhibits no distinct Curie point attributable to magnetite while sites such as 26 and 27 (Fig. 4) exhibit a distinct Curie point due to this mineral residing in a minor fraction of the initial saturation remanence. These sites undergo dramatic changes (type D curves, Table 2) during heating or cooling with production of a phase or phases with much higher saturation magnetisations; this probably represents a transformation of sulphides to magnetite because higher Curie points due to hematite are not observed.

Quite different curves, designated type B in Table 2 are observed at four sites and show a significant fall in moment on cooling. This might in part be explained by production of a hematite phase (with lower saturation magnetisation) from ilmenite during cooling, but this cannot be the complete explanation because a lower Curie point is sometimes evident from the cooling curve (Table 2). It is noteworthy that all the sites with moderate to good magnetic stability are associated with type A or B thermomagnetic curves. An observation of minor importance from the curves at three sites is the increase in J_S on heating through a temperature range at about 200°C (''x'' points, Table 2); a ferrimagnetic

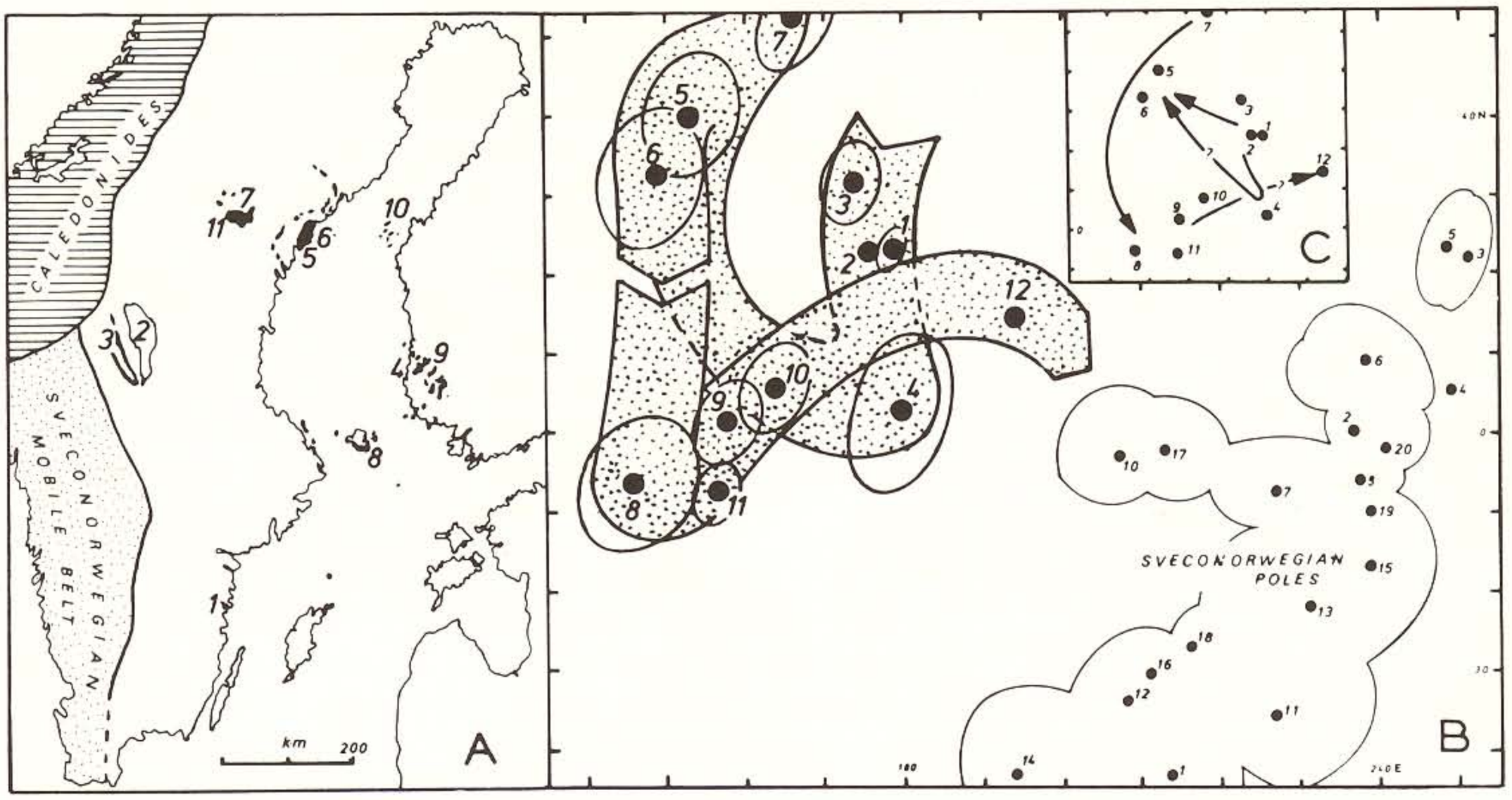


Fig. 5. □ A. The regional location of the palaeomagnetic studies listed in Table 3 and yielding magnetisations post-dating the Svecofennian tectonism but predating the Sveconorwegian tectonism of south-western Sweden. □ B. Apparent polar wander path defined by the results listed in Table 3 which are plotted as large circles; 95% confidence ovals are also indicated where appropriate. The poles plotted as small circles represent magnetisations assigned to the interval ca. 1100–700 Ma and derived in part from the Sveconorwegian terrain. These poles are plotted and listed in the order given by Patchett & Bylund (1977). □ C. The time sequence of the poles given in Table 3 with the main uncertainties in the sequence indicated.

phase is being produced by heating in these cases and this probably represents formation of magnetite from a hydrous phase such as goethite or lepidocrocite.

Interpretation of the palaeomagnetic poles

The palaeomagnetic results from the Ragunda region may be interpreted in the context of other palaeomagnetic studies from the Fennoscandian Shield relating to the interval between uplift and cooling of the regional Svecofennian rocks, and prior to the intrusion of the post-Jotnian dolerites (1270–1210 Ma, Patchett et al. 1978) and the development of the Sveconorwegian mobile belt (<1100 Ma) at the south-western margin of the Shield (Fig. 5A). The present palaeomagnetic results relevant to this interval are summarised in Table 3. Pole 1 from the syntectonic Loftahammar gabbro (Poorter 1976) is probably the oldest of these poles with the age of the magnetisation lying somewhere between the regional tectonism dated by a Rb–Sr whole-rock isochron age at 1658 ± 35 Ma and regional uplift and cooling dated by mineral ages in the range 1475–1470 Ma. In Table 3 ages have been recalculated using the new radionucleiide decay constants given by Steiger & Jäger (1977). Palaeomagnetic poles from post-tectonic sedimentary/volcanic successions include pole 3 from the Dalarna basalts (Mulder 1971) and pole 4 from the Satakunta Sandstone of Finland (Neuvonen 1974). The Ragunda complex yields a significantly different palaeomagnetic pole from the other units listed; this probably reflects a real difference in age between this body and the other rock units listed in Table 3. The nearby and slightly older Nordingrå Rapakivi suite has a more complex history of magnetisation (Piper 1979) due to an extensive heating and partial remagnetisation of much of the intrusion by the later post-Jotnian dolerite igneous activity. These post-Jotnian dolerites have been dated by the recent Rb–Sr studies of Patchett et al. (1978) which yield a range of ages mostly lying between 1247 and 1213 Ma. They are all of one magnetic polarity (Piper 1979) and yield a well-defined group of poles separated by 40–50° from the Ragunda and Nordingrå poles (Fig. 5B). The palaeomagnetic pole from the post-Ragunda dolerite dykes differs from the pole for the post-Jotnian dolerites of Västernorrland County 50–100 km further to the east (Larson & Magnusson 1976; Piper 1979)

Table 3. Summary of 1600–1200 Ma palaeomagnetic data for the Fennoscandian Shield.

Rock unit	Radiometric[a] age	Palaeomagnetic sample: sites/samples	*dp/dm*	Palaeomagnetic pole
1. Loftahammar Gabbro	1658–1470	5/26	2/5	179°E, 23°N
1. Upper Dala volcanics	1634–1459	2/11	–	176°E, 23°N
3. Dalarna basalts	–	2/11	–	174°E, 32°N
4. Satakunta Sandstone	1431–1390	6/10	8/12	180°E, 3°N
5. Nordingrå Gabbro[b]	> 1385	13/100	7/12	153°E, 40°N
6. Nordingrå Granite	1385	–/28	8/14	149°E, 33°N
7. Ragunda Complex	1293	19/111	6/9	165°E, 54°N
8. Märket dolerites	1270	7/9	7/11	146°E, 6°S
9. Satakunta dolerites	1236, 1225	15/18	3/5	158°E, 2°N
10. Vaasa dolerites	1270, 1225	15/15	3/6	164°E, 7°N
11. Post-Jotnian dolerites of central-east Sweden[c]	1254	48/322	2/3	157°E, 7°S
12. Post-Ragunda dolerites	–	12/64	8/15	194°E, 15°N

[a] Published ages recalculated to conform with new decay constants (Steiger & Jäger 1977).
[b] Result based on a.f. demagnetisation of sites showing a magnetisation predating the post-Jotnian dolerites.
[c] Mean of several separate studies given in Piper (1979). *Dp* and *dm* are the semi axes of the ovals of confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively; these are only given where the result is based on an adequate statistical sample.

and dated 1254 Ma by the K–Ar isochron method (Welin & Lundqvist 1975). The magnetic properties of the post-Ragunda dykes are also quite unlike the post-Jotnian dolerites which possess a very stable high coercivity remanence residing only in titanomagnetite. It is therefore inferred that they represent different episodes of dolerite intrusion; in the absence of radiometric control for the post-Ragunda dykes, the nature of this age difference is uncertain, but it is noteworthy that this pole lies between the palaeomagnetic poles from post-Jotnian dolerites (ca. 1270–1210 Ma) and poles from the Sveconorwegian mobile belt with uplift magnetisations of 1100 Ma and younger in age (Patchett & Bylund 1977).

The Middle–Late Proterozoic palaeomagnetic poles from the Fennoscandian Shield summarised in Table 3 are plotted in the form of an a.p.w. curve in Fig. 5B and the uncertainties in the form of this curve are indicated diagrammatically in Fig. 5C. There is a real difference of 20–30° between poles 1 and 3 (ca. 1700–1500 Ma) and the Nordingrå and Ragunda poles 5 and 7 (ca. 1390–1290 Ma), but it is not completely certain how the palaeomagnetic pole 4 from the Satakunta Sandstone of Finland should be incorporated into this curve. Although a general age of about 1300 Ma had been accepted for this formation, Neuvonen (1973) noted the uncertainty surrounding this age; Poorter (1976) quotes an age range of ca. 1430–1390 Ma on the basis of age evidence relating to the interval of Jotnian sedimentation within the Fennoscandian Shield. Thus this pole probably predates the magnetisations of the Ragunda and Nordingrå intrusions but it may be younger.

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